

Intelligent OCR for Scanned Documents

French — English — Arabic

Technical note: rationale for the OCR engine selection

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1 Context and Objectives

The project addresses the need to digitise and automatically process **scanned** documents: administrative forms, reports and handwritten material, written in **French**, **English** or **Arabic**.

Classical OCR engines (Tesseract, for example) detect *character shapes* with no understanding of context, layout or language. They degrade sharply on noisy scans, mixed scripts, tables and handwriting — precisely the cases encountered here.

By contrast, a **Vision-Language Model** (VLM) reads an image and produces text within a *single* neural network. It is this joint understanding of image and language that makes difficult documents tractable.

Objectives.

- Faithfully transcribe multilingual scanned documents (FR / EN / AR, including right-to-left script).
- Preserve the document **structure** (headings, tables, checkboxes), not merely the raw text.
- Guarantee **confidentiality**: the reference document used for testing is a strictly confidential safeguarding form containing names and medical details. No document may leave the machine.
- Run on the hardware actually available: a laptop fitted with an **NVIDIA RTX 4050 (6 GB VRAM)**.
- Incur no recurring cost: no per-page billing, no quota.

2 Functional Scope

In scope.

- **Inputs**: images and scanned documents (individual pages).
- **Languages**: French, English, Arabic — including mixed FR/AR documents and non-Latin right-to-left scripts.
- **Content handled**: dense text, tables, structured forms, checkboxes, handwriting.
- **Output**: structured text in **Markdown**, preserving the hierarchy and tables of the source document.
- **Execution modes**: local engine by default, second opinion (quality assurance) on a larger model, optional cloud engine.
- Full **offline operation**.

Out of scope.

- Cloud processing as the default: ruled out for data-governance reasons.
- Business validation of the extracted content: the system transcribes; it neither interprets nor validates medical information.
- High-throughput batch processing: the target is the individual document, at roughly 30 seconds per page.

3 Overall Architecture

The architecture rests on one guiding principle: **local-first**. The nominal engine runs entirely on the user's machine; the cloud is only a fallback option.

Processing chain.

1. **Ingestion**: loading the image or scan.
2. **Preparation**: image normalisation and construction of the transcription prompt.
3. **VLM inference**: reading the image and generating the text with the vision-language model, served locally.
4. **Structuring**: Markdown formatting of the output (headings, tables, lists).
5. **Second opinion (optional)**: a further pass on a larger model for comparison and quality control.
6. **Delivery**: hand-off of the transcribed document.

Role	Model	Where it runs
Default engine	Qwen3-VL 4B (instruct) — `qwen3-vl:4b-instruct`	Local, via Ollama, on the GPU
Second opinion (QA)	Qwen3-VL 8B (instruct) — `qwen3-vl:8b-instruct`	Local, via Ollama
Optional cloud	Llama 4 Scout (vision)	Groq API — currently unavailable

Table 1: Division of roles between the engines

Engines involved.

4 Detailed Technical Architecture

4.1 Local inference service

The models are served by **Ollama**, which exposes a local API and manages loading the weights onto the GPU. No outbound network call is required for nominal operation.

4.2 Memory constraint and GPU residency

The RTX 4050 provides **6 GB of VRAM**. The 4B model occupies about **3.3 GB** and therefore stays *fully resident* on the GPU. Larger models spill over to the CPU, which severely degrades response times.

4.3 The `-instruct` suffix is mandatory

A decisive technical point: the base `qwen3-vl` tags run in *thinking* mode. They spend the entire token budget on internal reasoning and return an **empty transcription**. Only the `-instruct` variant transcribes the image content directly.

4.4 Context-window tuning

Setting the `num_ctx` parameter to **8192 tripled the speed** compared with the naive initial setting, bringing processing down to roughly **30 seconds per page** on the target hardware.

4.5 Cloud path (optional)

A much larger remote engine — **Llama 4 Scout** (17B, vision) via the **Groq** API — was integrated as a benchmark and fallback for cases willing to trade confidentiality for maximum accuracy.

5 Validation and Known Limitations

5.1 Measurement protocol

The comparison was run on the most demanding case in the corpus: a **dense handwritten form** containing tables and checkboxes. The chosen criterion is transcription **completeness**, supplemented by processing time.

Model	Completeness	Duration	Observation
Qwen3-VL 4B instruct	10/10	59 s	Fully resident on the GPU
Qwen3-VL 8B instruct	9/10	135 s	Spilled to CPU (6 GB VRAM insufficient)

Table 2: Results measured on the reference handwritten form

The result is counter-intuitive but unambiguous: on this machine the 4B model is both **faster and more complete** than the 8B. The choice is therefore not assumed, it is **measured**.

5.2 Known limitations

- **Cloud path unavailable.** Since mid-2026, Groq has removed all Llama 4 vision models from general access: that engine currently returns a `404`. This unavailability retrospectively vindicates the local-first choice — the local engine depends on no third party.
- **Hardware dependency.** Performance is tied to the presence of a GPU and its VRAM; on a machine without a GPU, processing times become prohibitive.
- **Scope of the measurement.** The figures above come from one reference document on one machine; they indicate a trend, not a universal guarantee.
- **Fidelity not guaranteed at 100%.** Given that the data is personal and medical, **human proofreading** of the transcription remains necessary.
- **Arabic handwriting.** It remains the hardest case in the chain and accounts for most of the residual errors.

6 Technology Choices

6.1 Why a VLM rather than classical OCR

Classical OCR recognises glyphs; a VLM *understands* the page. That understanding of context, layout and language is what makes it possible to handle degraded scans, mixed scripts, tables and handwriting.

6.2 Why the Qwen3-VL family

- **Strong multilingual performance, Arabic included.** The project targets French, English and Arabic (right-to-left). Qwen models are trained heavily on multilingual and

Arabic data, and rank among the best open VLMs on non-Latin scripts — confirmed on the mixed FR/AR documents in the corpus.

- **Built for document tasks.** They handle dense text, tables and structured forms, and produce structured **Markdown** output: exactly what the reference form required, with its tables and checkboxes.
- **Open weights, local execution.** Usable through Ollama with no licence fee.

6.3 Why the 4B size, and why the *instruct* variant

- **4B fits the hardware.** About 3.3 GB, meaning full residency within the 6 GB of VRAM on the RTX 4050.
- **Measured, not assumed.** 10/10 in 59 s for the 4B against 9/10 in 135 s for the 8B (see Section 5).
- **The `-instruct` tag is imperative.** Without it, the model reasons instead of transcribing and returns an empty output.

6.4 Why local (Ollama) is the default engine

- **Confidentiality.** The test document is a strictly confidential safeguarding form with names and medical details. Local processing means *no document ever leaves the machine*: the decisive factor for data governance.
- **No cost, no limits, offline.** No per-page API fees, no rate limits, no need for an internet connection.
- **Fast enough on real hardware.** Roughly 30 seconds per page after tuning the context window.

6.5 Why Groq is kept as an option

This remote engine serves as a **benchmark** and a **fallback** towards a substantially larger model, for cases that would accept trading confidentiality for maximum accuracy. Its withdrawal by the provider does, however, illustrate the risk of depending on a third-party service.

7 Project Structure

The breakdown below reflects the separation between the OCR engines, the processing chain and evaluation.

Component	Responsibility
Engine layer	Common abstraction over the OCR engines; local implementation (Ollama) and cloud implementation (Groq).
Processing chain	Image ingestion, preparation, engine invocation, Markdown structuring.
Configuration	Model selection, inference parameters (including `num_ctx`), choice of default engine and second-opinion engine.
Evaluation	Corpus of reference documents, comparison protocol, completeness and duration records.
Outputs	Markdown transcriptions produced, kept locally.

Table 3: Functional breakdown of the project

This breakdown makes it possible, in particular, to swap engines — local or cloud — without touching the processing chain.

8 Development Planning

Phase	Title	Content
1	Study and framing	Analysis of the multilingual requirement, assessment of the limits of classical OCR, identification of the confidentiality and hardware constraints.
2	Model selection	Review of open VLMs, selection of the Qwen3-VL family for its multilingual and document-handling performance.
3	Local prototyping	Bringing Ollama into service, first transcription trials, identification of the *thinking*-mode problem and switch to the `-instruct` variants.
4	Optimisation	Context-window tuning (`num_ctx` = 8192), speed tripled, verification of GPU residency.
5	Comparative evaluation	4B versus 8B on the reference handwritten form; measurement of completeness and timings.
6	Cloud integration	Addition of the Groq / Llama 4 Scout engine as a fallback; observation of its withdrawal by the provider.
7	Consolidation	Fixing the default engine, setting up the second opinion, documenting the known limitations.

Table 4: Development phases

9 Conclusion

The system uses **Qwen3-VL-4B-Instruct**, an open vision-language model served locally via **Ollama**, as its default OCR engine.

The choice rests on three converging arguments: its multilingual performance, notably in Arabic; its ability to handle documents, tables and handwriting; and — critically — the fact

that it runs **fully offline on a 6 GB laptop GPU**, keeping confidential documents on the device.

The 4B size was selected over the 8B after **direct measurement**, which showed it to be both faster and more complete on the target hardware. A cloud engine (Groq / Llama 4) is retained as a higher-capacity fallback; vision access having since been withdrawn by the provider, that unavailability underscores precisely the value of the *local-first* design.